at the University of Stockholm. In the 1890s, he decided that past ice ages might have been caused by fewer volcanic eruptions pumping gases such as carbon dioxide into the atmosphere. These gases retain heat, so reducing them would, he argued, cool down Earth. He also noted that burning fossil fuels would increase these gases and make the Earth warm up; in this way, he linked human activity with rising global temperatures, paving the way for modern concerns about climate change.

ANNIE JUMP CANNON

1863–1941

American astronomer Annie Jump Cannon was the 20th century's leading authority on the spectra of stars. Born in Delaware, she studied physics and astronomy at Wellesley College and joined the Harvard College Observatory in 1896 as one of several women hired to process astronomical data. She pioneered the classification of stars with her 1901 system, which laid the foundations for the Harvard Spectral Classification system. Over 44 years, she classified 350,000 stars.

HENRIETTA SWAN LEAVITT

1868–1921

While studying at Radcliffe College in Cambridge, Massachusetts, Henrietta Swan Leavitt became interested in astronomy. She went on to examine the luminosity of stars from thousands of photographic plates at the Harvard College Observatory. She saw that Cepheid variable stars (pulsating stars) showed a regular pattern of brightness. Her work was crucial for measuring the distance between the Earth and other galaxies. Leavitt discovered more than 2,400 variable stars and four novae (bright, transient interactions between stars). She also developed a standard for photographic measurements, now called the Harvard Standard. Her work was largely unrecognized in her lifetime.

HARRIET BROOKS

1876–1933

Born in Ontario, Canada, Harriet Brooks graduated from McGill University in 1901 and became Canada's first female nuclear physicist. She studied under J. J. Thomson in Cambridge, UK, and Ernest Rutherford in Canada and worked in Marie Curie's laboratory in Paris from 1906. She discovered that one element could change into another through nuclear decay. Several of her findings were only attributed to her after her death.

SRINIVASA RAMANUJAN

1887–1920

Born in Madras (now Tamil Nadu), India, Srinivasa Ramanujan made major contributions to mathematical analysis and number theory despite little formal training. He was invited to Cambridge in 1913, after sending mathematician G. H. Hardy a letter and 120 mathematical theorems. Ramanujan was awarded a Bachelor of Science degree in 1916, and soon after became the second Indian to be elected a Fellow of the Royal Society. He contracted tuberculosis in 1916 and returned to India 2 years later, but died in 1920.

ERWIN SCHRODINGER

1887–1961

Born in Vienna, Austria, in 1887, Erwin Schrödinger studied physics at the University of Vienna. He moved to Germany, then to the University of Zurich, Switzerland, where he carried out his most important work, immersing himself in the emerging field of quantum physics. Schrödinger's wave equation allowed a new understanding of how some particles behave. Rather than imagining electrons as orbiting the atom's nucleus, arranged in shells and subshells, the wave equation shows that orbitals, shells, and subshells are actually "clouds" of probability that tell us how likely it is for a particular electron to be found in a specific position. Schrödinger's equations of wave mechanics changed the way we see the world and formed the basis for today's quantum mechanics.

RONALD FISHER

1890–1962

British statistician and geneticist Ronald Fisher pioneered the application of statistics to scientific experimentation. In 1918, he published a paper that illustrated the use of statistical tools to reconcile what were apparent inconsistencies between Charles Darwin's ideas of natural selection and the recently rediscovered experiments of botanist Gregor Mendel. Fisher was knighted in 1952.

HAROLD UREY AND STANLEY MILLER

1893–1981; 1930–2007

In 1953, American chemists Harold Urey and Stanley Miller simulated early Earth in the laboratory with electrical sparks to imitate lightning. They used a closed series of connected glass flasks, sealed from the atmosphere, and placed in them water and a mixture of gases thought to have been present in Earth's primitive atmosphere—hydrogen, methane, and ammonia. They showed that with enough heat and energy, simple life-giving, carbon-based compounds could be produced.